BTS–IV(SS)–02.20–0073 Reg. No. [ ]

A

B.Tech. Degree IV Semester Special Supplementary Examination February 2020

ME 15-1404 APPLIED THERMODYNAMICS

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) Describe thermodynamic equilibrium of a system.

(b) Write equivalence amongst different temperature scales. Also write brief note on each of them.

(c) Show that for a polytropic process.

$$Q=\left(\frac{\gamma-n}{\gamma-1}\right)W.$$

(d) Derive the expression for the following:

(i) Internal energy of steam

(ii) Entropy of wet steam

(iii) Entropy of superheated steam

(e) Water at 120°C with a quality of 25% has its temperature raised 20°C in a constant volume process. What is the new quality and pressure?

(f) Compare the efficiency and specific steam consumption of ideal Carnot cycle with Rankine cycle for given pressure limits.

(g) What is meant by Mollier diagram? Explain.

(h) Explain supersaturated flow in steam nozzle.

(i) Explain Amagat's Law.

(j) A spherical helium balloon of 10 m in diameter is at ambient T and P, 15°C and 100 kPa. How much helium does it contain? It can lift a total mass that equals the mass of displaced atmospheric air. How much mass of the balloon fabric and cage can then be lifted?

PART B

(4 × 10 = 40)

II. (a) Derive steady flow energy equation. Apply SFEE to compressor, nozzle and throttling device. (4)

(b) Three tanks of equal volume of 4 m^3 each are connected to each other through tubes of negligible volume and valves in between. Tank A contains air at 6 bar, 90°C, tank B has air at 3 bar, 200°C and tank C contains nitrogen at 12 bar, 50°C. Considering adiabatic mixing determine , (6)

(i) the entropy change when valve between tank A and B is opened until equilibrium

(ii) the entropy change when valves between tank C, tank A and tank B are opened until equilibrium.

Consider R_{Air} = 0.287 kJ/kg. K, γ_{Air} =1.4, $R_{Nitrogen}$ = 0.297 kJ/kg. K and $\gamma_{Nitrogen}$ = 1.4.

OR

III. (a) Explain Helmholtz function, Gibbs function and Maxwell relations. Discuss their significance. (4)

(b) In a steam turbine the steam enters at 50 bar, 600°C and 150 m/s and leaves as saturated vapour at 0.1 bar, 50 m/s. During expansion, work of 1000 kJ/kg is delivered. Determine the inlet stream availability, exit stream availability and the irreversibility. Take dead state temperature as 25°C. (6)

IV. (a) Describe Regenerative cycle and compare it with simple Rankine cycle. (5)

(b) A power plant produces steam at 3 MPa, 600°C in the boiler. It keeps the condenser at 45°C by transfer of 10 MW out as heat transfer. The first turbine section expands to 500 kPa and then flow is reheated followed by the expansion in the low pressure turbine. Find the reheat temperature so the turbine output is saturated vapour. For this reheat find the total turbine power output and the boiler heat transfer. (5)

OR

V. (a) Explain Mercury - Water binary vapour cycle. (4)

(b) In a steam turbine installation running on ideal Rankine cycle steam leaves the boiler at 10 MPa and 700°C and leaves turbine at 0.005 MPa. For the 50 MW output of the plant and cooling water entering and leaving condenser at 15°C and 30°C respectively determine: (6)

(i) the mass flow rate of steam in kg/s
(ii) the mass flow rate of condenser cooling water in kg/s
(iii) the thermal efficiency of cycle
(iv) the ratio of heat supplied and rejected (in boiler and condenser respectively).

Neglect K.E. and P.E. changes.

VI. (a) Explain the effect of friction on nozzle. (4)

(b) Determine the mass flow rate of steam through a nozzle having isentropic flow through it. Steam enters nozzle at 10 bar, 500°C and leaves at 6 bar. Cross-section area at exit of nozzle is 20 cm^2. Velocity of steam entering nozzle may be considered negligible. Show the process on h-s diagram also. (6)

OR

VII. A pass out turbine is fed with steam at 20 bar, 250°C. The pass out steam pressure is 5 bar and the exhaust pressure is 0.075 bar. In this installation the total power required is 4500 kW and heat load is 15000 kW. The efficiency of HP and LP turbine stages is 0.8 in each stage. Considering the reduction in total steam consumption to result in steam consumption rate of 10 kg/s, determine the new power output if the heat load remains same. Also assume the nozzle control governing and throttle governing at the inlets of HP turbine and LP turbine respectively. (10)

VIII. (a) Derive an expression for the apparent molecular mass of a mixture of gas. (5)

(b) Determine the pressure of 5 kg carbon dioxide contained in a vessel of 2 m^3 capacity at 27° C, considering it as (5)

(i) perfect gas
(ii) real gas

OR

IX. (a) Explain volumetric and gravimetric analysis of gas mixtures. (5)

(b) The molar analysis of the gaseous products of combustion of a certain hydrocarbon fuel is CO2, 0.08; H2O, 0.11; O2, 0.07; N2, 0.74. Determine: (5)

(i) the apparent molecular weight of the mixture.
(ii) the composition in terms of mass fractions (gravimetric analysis).

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